

1 Write a Python program using lambda, map, filter, and reduce on a list of numbers.

AIM

To write a Python program that applies lambda functions along with map, filter, and reduce on a list of numbers.

ALGORITHM

1. Start the program.
2. Create a list of numbers.
3. Use map() with a lambda function to square each number.
4. Use filter() with a lambda function to keep only even numbers.
5. Use reduce() with a lambda function to calculate the sum of all numbers.
6. Display results.
7. End program.

CODE

```
from functools import reduce
```

```
# Initial list
```

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```
print("Original List:", numbers)
```

```
# 1. Using map() with lambda - square each number
```

```
squares = list(map(lambda x: x**2, numbers))
```

```
print("Squares using map:", squares)
```

```
# 2. Using filter() with lambda - filter even numbers
```

```
evens = list(filter(lambda x: x % 2 == 0, numbers))
```

```
print("Even numbers using filter:", evens)
```

3. Using reduce() with lambda - sum of all numbers

```
sum_all = reduce(lambda x, y: x + y, numbers)
```

```
print("Sum using reduce:", sum_all)
```

OUTPUT

Original List: [1, 2, 3, 4, 5, 6, 7, 8, 9]

Squares using map: [1, 4, 9, 16, 25, 36, 49, 64, 81]

Even numbers using filter: [2, 4, 6, 8]

Sum using reduce: 45

CONCLUSION

This program demonstrates:

- Lambda functions → anonymous functions for quick operations.
- map() → applies a function to each element (squares).
- filter() → selects elements based on a condition (evens).
- reduce() → reduces a list to a single value (sum).

2 Write a Python program to demonstrate the use of zip() and enumerate().

AIM

To write a Python program that shows how and can be used to handle multiple sequences and iterate with indexes.

ALGORITHM

1. Start the program.
2. Create two lists: one for student names and one for their marks.
3. Use to combine both lists into pairs.
4. Display the combined results.

5. Use to iterate through a list with index numbers.
6. Display the indexed results.
7. End program.

CODE

```
# Demonstrating zip() and enumerate()
```

```
# Lists of students and marks
```

```
students = ["Yogin", "Amit", "Priya", "Sneha"]
```

```
marks = [85, 90, 78, 92]
```

```
# Using zip() to combine students and marks
```

```
print("Using zip():")
```

```
for name, score in zip(students, marks):
```

```
    print(f"{name} scored {score} marks")
```

```
# Using enumerate() to display index with values
```

```
print("\nUsing enumerate():")
```

```
for index, name in enumerate(students, start=1):
```

```
    print(f"Student {index}: {name}")
```

OUTPUT

```
Using zip():
```

```
Yogin scored 85 marks
```

```
Amit scored 90 marks
```

```
Priya scored 78 marks
```

```
Sneha scored 92 marks
```

```
Using enumerate():
```

Student 1: Yogin

Student 2: Amit

Student 3: Priya

Student 4: Sneha

CONCLUSION

This program demonstrates:

- → combines multiple sequences element-wise (students with their marks).
- → provides index numbers while iterating through a sequence.